RESTRICTED

ABS LineCheck Communication Specification

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part name: ABS Unit Assy

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F	10.07.13	English description was added.(英語併記対応)	M.Takahashi	Y.Takahara	N.Endo
E	10.04.20	J68N/J68CC/J68T/J64T追加、組織名変更. Cotrol Mask コメント追加。	内藤	高原	遠藤奈
			車シ開部 減速回生TM		
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C	'09.05.29	J74G DSC追加	高橋	高原	尾中
B	08.06.02	駆動用ControlMaskﾃﾞｰﾀ修正	高橋	山下	尾中
A	08.03.07	Initial Document setup			尾中

Table of Contents

		Page
0.車種ﾗｲﾝﾁｪｯｸ実施内容一覧表 ~~0.line check execution contents~~ G	･･････････････････････････････	3
1.ABS LineCheck Communication Specification	･･････････････････････････････	4
2.General Flow Chart	･･････････････････････････････	8
3.DIAGCLR01	･･････････････････････････････	9
4.BRKCHK01	･･････････････････････････････	10
5.SPCHK01	･･････････････････････････････	12
6.FCCHK01	･･････････････････････････････	14
7.Detection for communication error	･･････････････････････････････	16
8.Revision	･･････････････････････････････	17
Reference) Bit set for Valve control	･･････････････････････････････	18

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車種
Project

J23A(
J2

J

B
C
D
E

(*9)Security Access
Tester may send the request of Security Acess(527). Then ABS will send NRC 535.

1.ABS LineCheck Communication Specification (the spec. for DSC is the same as ABS)

1.the confirmation of the connection with the diagnosis connector

The operator connect the connector of the tester with the diagnosis(J1962) connector of the vehicle.
The tester monitor the A terminal voltage of J1962 connector applied +B(12 volts) to detect connecting with the connector.

(Note) The tester shall stop the CAN communication function until the completion for the connection of the connector.

2.the judgement for the Ignition key is in the ON position.

The operator turn on the engine after connecting the connector.
The tester start to receive the CAN signal after the judgement of connection.
The tester decide the ignition key is in the ON position when the intermodule CAN ID data is received.
(note) CAN ID is decided the project each.

3-1.the condition for the start of ABS line check inspection

The tester start to ABS line check inspection after the decision of the ignition key is in the ON position.

3-1-1. The condition for permission of the performance of ABS Line Check.

The tester send the entry signal for the extended session to ABS module.The communication spec. is shown in belows.
(note) If ABS module does not receive the data from the tester more than 5 sec after the entry to the extended session($03) once,
ABS module judge the time-out and return to the normal mode.Therefore ABS module has to transmit the signal of "Tester presense($3F)"
so that prevention ABS module to return the default session.
ABS module perform flashing the ABS warning light in case of entering to the extended session.
The operator should check that the ABS warning light is flashing to confirm of performing of the ABS line check.

It is necessary to be satisfied the following condition.If not be satisfied,the tester judge to "communication error" and handle to "NG".
1.the master vehicle speed is less than 10 km/h.
2.ABS does not detect the failure.(the present DTC does not exist.)

3-1-2. The command for enrty to the ABS line check is shown belows.

(Summary)
The tester read the present DTC after the past DTC is cleared, and then if the number of DTC is nothing,the performance of the ABS line check is permitted.

1. tester->ECU **Request command of Clear DTC**

A data byte	Hex value	
#1	$14	SID
#2	$FF	Data Parameter all Group of DTC
#3	$FF	
#4	$FF	

2. ECU->tester

Positive Response (ABS ECU accept the request.)

A data byte	Hex value	
#1	$54	SID

Negative Response (ABS ECU can't accept the request.)

A data byte	Parameter name	Hex value
#1	negative Response Service Id	7F
#2	request Service Id	00-3F,80-BF
#3	negative Response Code	00-FF

#3 :negative Response Code(NRC)
Negative Response Code(NRC) is the detail description for the negative Response.

negativeResponseCode

Hex value	Description
11	service Not Supported
12	sub-function Not Supported
21	busy
22	condition Not Correct
31	request Out of Range
33	security Access Denied
35	invalidKey
78	responsePending
~~7E 7F~~	service Not Support In ActiveSession

G

3. tester->ECU **Request command of Read DTC**

A data byte	Hex value	
#1	$19	SID
#2	$02	sub-function $02 :report DTC by StatusMask
#3	$02	DTC Status Mask $02 :(status bit1=1) (report the present DTC)

4. ECU->tester

Positive Response (ABS ECU accept the request.)

A data byte	Hex value	
#1	$59	SID
#2	$02	sub-function $02 :report DTC by StatusMask
#3	$CA	DTC Status AvailabilityMask=$CA (In case that the Status Bit data ABS module supported are bit7,6,3,1)
#4	$**	DTC = $**,$@@、$##、$&&
#5	$@@	(report DTC adapted to DTC Status Mask $02 :(status bit1=1))
#6	$##	
#7	$&&	

5. tester-> ECU **Request commmand of the Extended Session entry from the tester to ABS module**

A data byte	Hex value	
#1	$10	SID
#2	$03	sub-function $03 :Extended diagnostic Session

ECU->tester

Positive Response (ABS ECU accept the request)

A data byte	Hex value	
#1	$50	SID
#2	$03	sub-function $03 :Extended diagnostic Session

3-1-3. The command for keeping the Extended Session

(purpose)

This command should be used to make the ABS ECU keep the Extended Session to prevent 5seconds time out.
When the communication between the tester and ECU will be stopped for more than 5seconds,then ECU will return from the Extended Session to the default session and all of the activation will be stopped.

tester->ECU **Service Request for Tester Present from the tester to ABS module**

A data byte	Hex value	
#1	$3E	SID
#2	$00	sub-function $00

ECU->tester

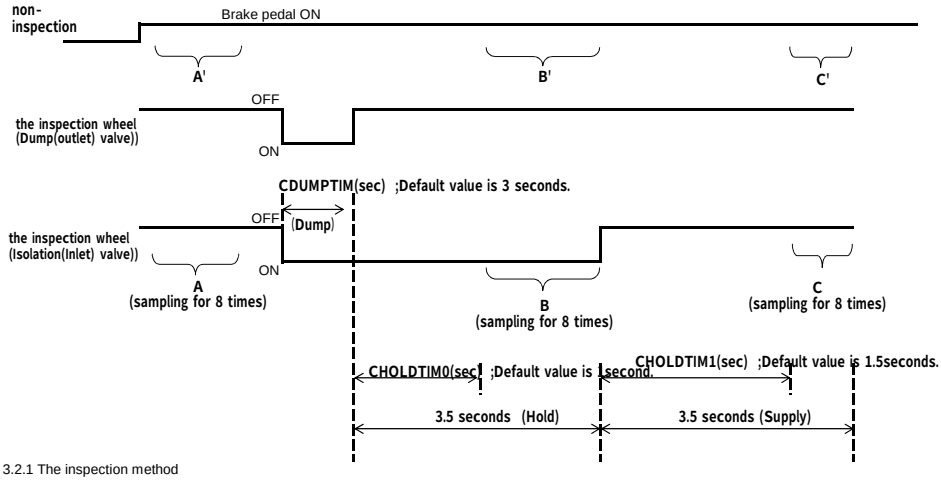
Positive Response (ABS ECU accept the request.)

A data byte	Hex value	
#1	$7E	SID
#2	$00	sub-function $00

3-2. The inspection spec. for the brake force of each wheel

The ABS line check perform the inspection for the brake force of each wheel.
The command to activate for the actuator(the solenoid valve and the pump motor) is shown as belows.
The condition for the activation are
(1)The master vehicle speed should be less than 10km/h.
and (2)ABS does not detect the failure(The present DTC does not exist).

The relation between the activation of the solenoid valves and the pattern is shown below.
The interface unit of the ABS line check only output according to the solenoid activation pattern as below.
The tester should inspect by comparison the actual brake force of the vehicle for the standard value.



3.2.1 The inspection method
The tester shall carry out this inspection by comparison the brake force of the inspection wheel and the non-inspection wheel.

3.2.2 The inspection criteria

αmin	<B	<αmax	comparison of the absolute pressure value (kgｆ)
βmin	<C/A	<βmax	comparison of the absolute value (%)
γmin	<B'/A'	<γmax	
γmin	<C'/A'	<γmax	

	min	max	comment
α	0	30	set the value of Switch (kgf)
β	60	150	set the value of Switch (%)
γ	80	150	set the fixed value of the software

The above value can only be determined by trials with production parts and online equipment.

3.2.3 The inspection command of the brake force is shown below.

1. tester->ECU **Request command of the actuator activation FOR ABS NC valve (refer to Reference1)**

A data byte	Hex value	
#1	$2F	:SID
#2	$2B	:DataIdentifier #1 ;2B2C BCM valves and pump motor
#3	$2C	
#4	$03	:Input/Output Control Parameter -> $03 ;I/O data is sent from Tester
#5	$data1	:$data1-$data4 :control state
#6	$data2	The data bit of the actuator should be activated is set.
#7	$data3	Dump:Isolation valve=ON,Dump valve=ON Supply:Isolation valve=OFF,Dump valve=OFF
#8	$data4	Hold:Isolation valve=ON,Dump valve=OFF Motor: is only activated in Dump mode .
#9	$data5	:$data5-$data8 :control mask
#10	$data6	~~the same data as $data1-$data4~~
#11	$data7	~~$data5-$data8=$FF, $F0, $FF, $FF~~
#12	$data8	

B

note) Either "same data as $date1-$date4" or "$date5-$data8=$FF,$F0,$FF,$FF" is applicable for J64F/J23A/J23AIPM.

E

2. ECU->tester

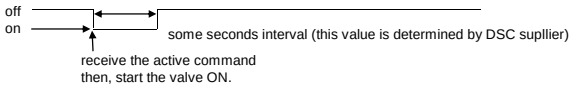
Positive Response (ABS ECU accept the request.)

A data byte	Hex value	
#1	$6F	:SID
#2	$2B	:DataIdentifier #1 ;2B2C BCM valves and pump motor
#3	$2C	
#4	$03	
#5	$data1	:$data1-$data4 :control state
#6	$data2	
#7	$data3	
#8	$data4	

(Reference1)
ABS active comand

ID Number	Identifier Name	size	Byte	Bit	Bit Description	
2B2C	BCM valves and pump motor	32	1	7	Rear Right Isolation Valve	0:off 1:on
				6	Rear Right Dump	0:off 1:on
				5	Rear Left Isolation Valve	0:off 1:on
				4	Rear Left Dump	0:off 1:on
				3	Front Right Isolation Valve	0:off 1:on
				2	Front Right Dump	0:off 1:on
				1	Front Left Isolation Valve	0:off 1:on
				0	Front Left Dump	0:off 1:on
			2	7		
				6		
				5		
				4		
				3	Front Right /Rear Left Traction Control Isolation valve 2	0:off 1:on
				2	Front Left / Rear Right Traction Control Isolation valve 1	0:off 1:on
				1	Front Right / Rear Left Traction Control Supply 2	0:off 1:on
				0	Front Left / Rear Right Traction Control Supply 1	0:off 1:on
			3	1	Valve Supply	0:off 1:on
				0	Pump Motor supply	0:off 1:on
			4		unused	

Note: When ABS module receive the active command of solenoid, the solenoid will be activated for some seconds.
After some seconds, the solenoid will be stopped to activate for protection.



Note)
The pump motor should be activated between the Dump mode.

3-2-4 Security Acces (SecurityAccess要の場合) (If it's necessary to Security Access) F

(purpose)

This command is used to unlock security guard for Activeation of solenoid valve.

(note)

If tester can't send any data to ABS for more than 5seconds, then ABS will exit from the Extended Session and return to the Default Session.
At this time, security status of ABS also return from UNLOCKED to LOCKED.
Therefore, the tester have to send some data to ABS within 5seconds to keep this mode.

(security algorithm)

Tester have to calcurate 3-bytes security key from 3-bytes security seed based on standard security algorithm.
Security seed is sent from ABS and different numbers each time.
Detailed security algorithm specification is separated. Fixed 5 bytes data defined in this is ' **B8 3F 24 7F FF**' for ABS.

tester-> ECU

Service Request for Security Seed from the tester to ABS

A data byte	Hex value	
#1	$27	SID
#2	$03	sub-function $03

ECU->tester

Positive Response (ABS accept the request)

A data byte	Hex value	
#1	$67	SID
#2	$03	sub-function $03
#3	XX	
#4	YY	Security seed from ABS
#5	ZZ	

tester-> ECU

Service Request for Security Key from the tester to ABS

A data byte	Hex value	
#1	$27	SID
#2	$04	sub-function $04
#3	xx	
#4	yy	Security key calcurated from seed.
#5	zz	

ECU->tester

Positive Response (ABS accept the request)

A data byte	Hex value	
#1	$67	SID
#2	$04	sub-function $04

3-3. The confirmation method for ON/OFF of the Brake pedal switch

The ABS line check shall inspect for the Brake pedal signal.
This inspection is that ON/OFF signal of the Brake pedal switch is right or not.
(1)the inspection for OFF failure of the Brake pedal switch
The tester confirm that the Brake pedal switch is ON condition when the inspection for the Brake force start.
If the Brake pedal switch is OFF condition,the tester keep to watch the Brake pedal condition.
If OFF condition of the Brake pedal switch continue to more than 5 seconds,the tester set NG for the inspection for the Brake force..

(2)the inspection for ON failure of the Brake pedal switch
The tester confirm that the Brake pedal switch is OFF condition at least one time when the inspection for the wheel speed sensor start.
If the Brake pedal switch is ON condition through the inspection for all of wheel speed sensor,the tester set NG for the inspection for the Front Right wheel speed sensor or the Rear Right wheel speed sensor.
3.3.1 The inspection command of the Brake pedal switch is shown below.

1. tester->ECU **Request command of PID monitor of Brake Pedal Switch**

A data byte	Hex value	
#1	$22	:SID
#2	$2B	:DataIdentifier #1 ;2B00 Brake Input Switch Status
#3	$00	

2. ECU->tester

Positive Response (ABS ECU accept the request.)

A data byte	Hex value	
#1	$62	:SID
#2	$2B	:DataIdentifier #1 ;2B00 Brake Input Switch Status
#3	$00	
#4	$data1	:$data1 bit7=Brake Pedal Switch(BLS) 0:OFF, 1:ON

3-4. The inspection spec. for the wheel speed sensor

The interface unit always watch the signal of all wheel speed sensors when the inspection for each wheel speed sensor is conducted.
The tester confirm that all of the wheel speed(km/h) are satisfied for the value in below table.

	FR	FL	RR	RL
the inspection for both of Front wheels	A	A	0～1	0～1
the inspection for both of Rear wheels	0～1	0～1	A	A

A:the inspection threshold for whee speed = 3 - 7 km/h

The rolling speed of the roller(Brake tester): 5.4 km/h ±2%

3.4.1 The inspection command of the wheel speed sensor is shown below.

1. tester->ECU **Request command of PID monitor of Wheel Speed**

A data byte	Hex value	
#1	$22	←:SID
#2	$@@	←:$@@,$&&:DataIdentifier
#3	$&&	$2B,$06 : FL wheel speed signal
		$2B,$07 : FR wheel speed signal
		$2B,$08 : RL wheel speed signal
		$2B,$09 : RR wheel speed signal

2. ECU->tester

Positive Response (ABS ECU accept the request.)

A data byte	Hex value	
#1	$62	:SID
#2	$@@	:$@@,$&&:DataIdentifier
#3	$&&	:$data1:Data of DataIdentufier
#4	$data1	

$2B,$06 : FL wheel speed signal
:$data1(1byte) :Data of FL wheel speed (km/h)
$2B,$07 : FR wheel speed signal
:$data1(1byte) :Data of FR wheel speed (km/h)
$2B,$08 : RL wheel speed signal
:$data1(1byte) :Data of RL wheel speed (km/h)
$2B,$09 : RR wheel speed signal
:$data1(1byte) :Data of RR wheel speed (km/h)

3-5. The inspection spec. for the clear of the fault code memory

The interface unit check for the EEPROM to confirm whether it is possible to access to the EEPROM of ABS module or not.
The EEPROM is cleared by this check.

3.5.1 The inspection method

The fault code memory of the ABS module is cleared.
After that,the remained fault code is checked.

3.5.2 The inspection command of the clear for the fault code memory is shown below.

1. tester->ECU **Request command of Clear DTC**

A data byte	Hex value	
#1	$14	SID
#2	$FF	Data Parameter all Group of DTC
#3	$FF	
#4	$FF	

2. ECU->tester

Positive Response (ABS ECU accept the request.)

A data byte	Hex value	
#1	$54	SID

3. tester->ECU **Request command of Read DTC**

A data byte	Hex value	
#1	$19	SID
#2	$02	sub-function $02 :report DTC by StatusMask
#3	$02	DTC Status Mask $02 :(status bit1=1) (report the present DTC)

4. ECU->tester

Positive Response (ABS ECU accept the request.)

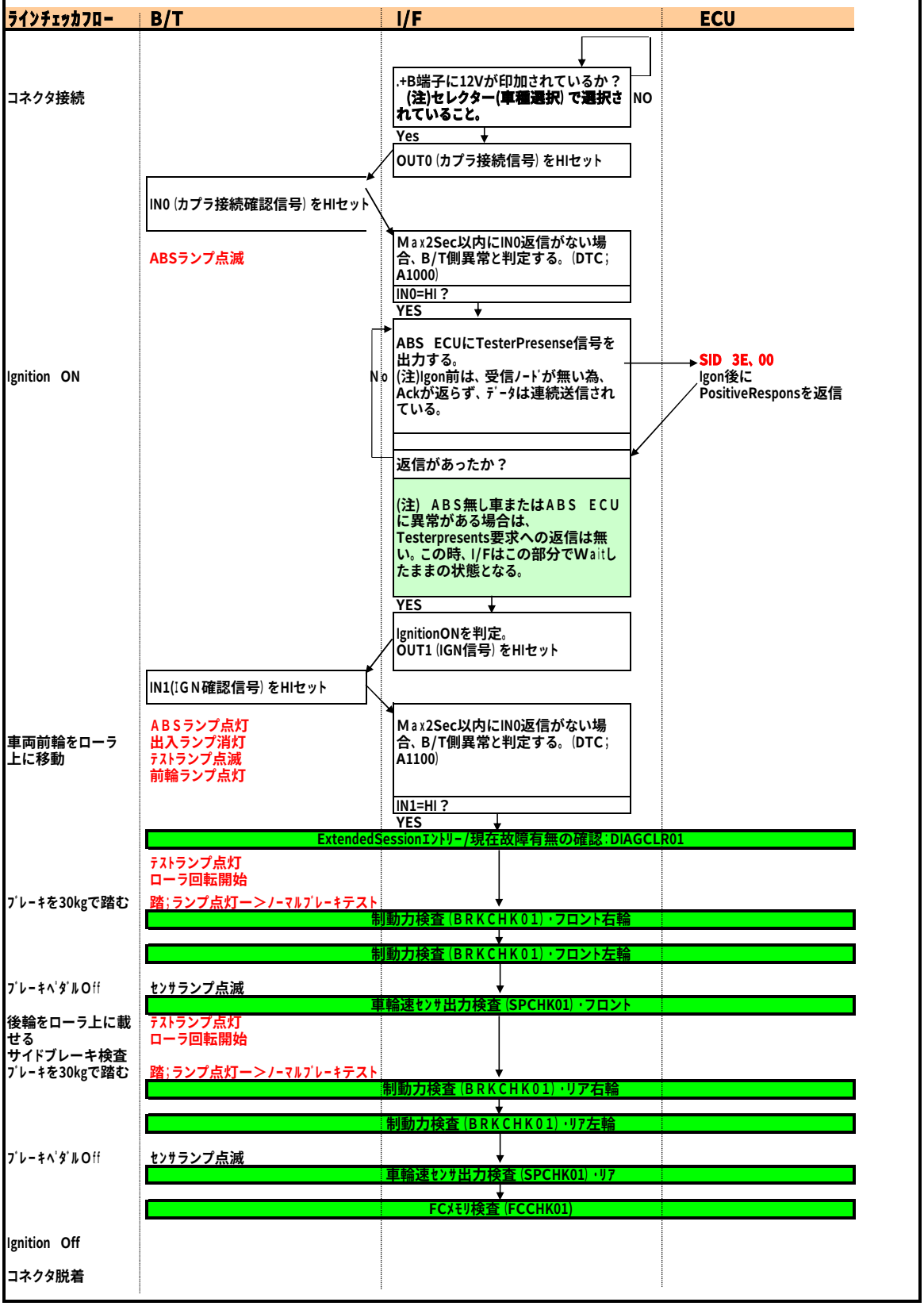
A data byte	Hex value	
#1	$59	SID
#2	$02	sub-function $02 :report DTC by StatusMask
#3	$CA	DTC Status AvailabilityMask=$CA (In case that the Status Bit data ABS module supported are bit7,6,3,1)
#4	$**	DTC = $**,$@@、$##、$&&
#5	$@@	(report DTC adapted to DTC Status Mask $02 :(status bit1=1))
#6	$##	
#7	$&&	

4.The completion of the ABS line checker

The ABS line checker finish the inspection in case of below condition.

i)ABS ECU does not receive the command from the tester for more than 5 seconds.
(This condition is included the situation of the ignition key OFF or the connector detached.)

2. 全体フロー

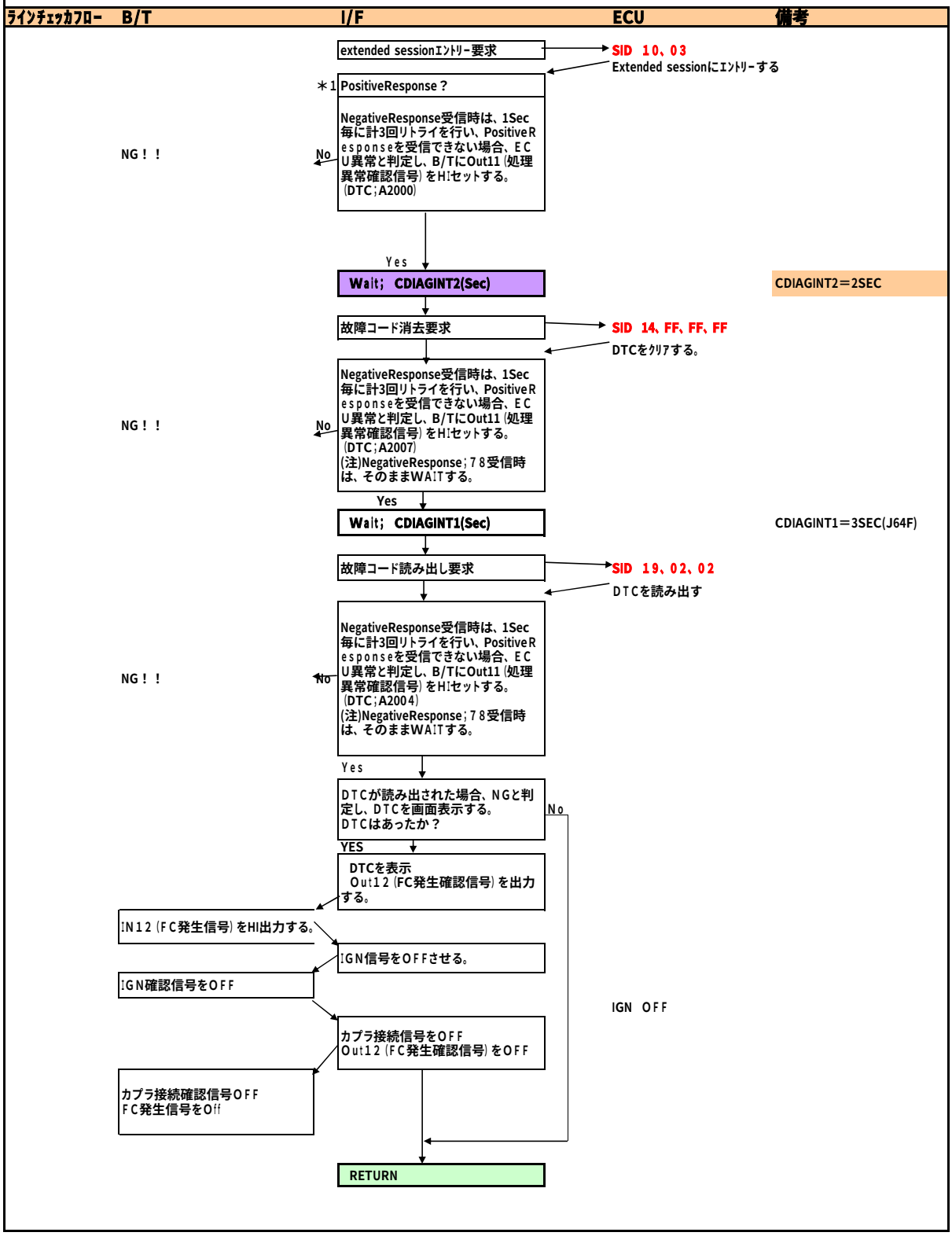

ﾗｲﾝﾁｪｯｸﾌﾛｰ
B/T
I/F
ECU
コネクタ接続
.+B端子に12Vが印加されているか？
(注)セレクター(車種選択)で選択されていること。
NO
Yes
OUT0(カプラ接続信号)をHIセット
IN0(カプラ接続確認信号)をHIセット
ABSランプ点滅
Max2Sec以内にIN0返信がない場合、B/T側異常と判定する。(DTC;A1000)
IN0=HI？
YES
Ignition ON
No
ABS ECUにTesterPresense信号を出力する。
(注)Igon前は、受信ﾉｰﾄﾞが無い為、Ackが返らず、ﾃﾞｰﾀは連続送信されている。
SID 3E、00
Igon後にPositiveResponsを返信
返信があったか？
(注) ABS無し車またはABS ECUに異常がある場合は、Testerpresents要求への返信は無い。この時、I/Fはこの部分でWaitしたままの状態となる。
YES
IgnitionONを判定。
OUT1(IGN信号)をHIセット
IN1(IGN確認信号)をHIセット
車両前輪をローラ上に移動
ABSランプ点灯
出入ランプ消灯
ﾃｽﾄランプ点滅
前輪ランプ点灯
Max2Sec以内にIN0返信がない場合、B/T側異常と判定する。(DTC;A1100)
IN1=HI？
YES
ExtendedSessionｴﾝﾄﾘｰ/現在故障有無の確認:DIAGCLR01
ﾃｽﾄランプ点灯
ローラ回転開始
ﾌﾞﾚｰｷを30kgで踏む
踏;ランプ点灯ー>ﾉｰﾏﾙﾌﾞﾚｰｷﾃｽﾄ
制動力検査(BRKCHK01)・フロント右輪
制動力検査(BRKCHK01)・フロント左輪
ﾌﾞﾚｰｷﾍﾟﾀﾞﾙOff
ｾﾝｻランプ点滅
車輪速ｾﾝｻ出力検査(SPCHK01)・フロント
後輪をローラ上に載せる
サイドブレーキ検査
ﾃｽﾄランプ点灯
ローラ回転開始
ﾌﾞﾚｰｷを30kgで踏む
踏;ランプ点灯ー>ﾉｰﾏﾙﾌﾞﾚｰｷﾃｽﾄ
制動力検査(BRKCHK01)・リア右輪
制動力検査(BRKCHK01)・ﾘｱ左輪
ﾌﾞﾚｰｷﾍﾟﾀﾞﾙOff
ｾﾝｻランプ点滅
車輪速ｾﾝｻ出力検査(SPCHK01)・ﾘｱ
FCﾒﾓﾘ検査(FCCHK01)
Ignition Off
コネクタ脱着

2. General flow

F English description was added.

Line checker flow	B/T	I/F	ECU
Connector connection		.Is 12V applied to +B terminal? **(Note) It shall be selected by selector (model choice).** NO Yes Set OUT0 (coupler connection signal) HI	
	Set IN0 (coupler connection confirmation signal) at HI.		
	Blink ABS lamp	When there is no IN0 reply within Max2Sec, it is assessed as B/T side error (DTC;A1000). IN0=HI ? YES	
Ignition ON		Output TesterPresense signal to ABS ECU. (Note) Before Igon, because there is no reception node, Ack does not return, and data is continuously sent. No	SID 3E、00 transmit PositiveRespons data after Igon.
		Is there response?	
		(Note)If there is the failure of ABS ECU or the vehicle model is no ABS,there is no response for TesterPresense signal.In this case,the I/F unit stays waiting at this situation.	
		YES Check Ignition ON. Set OUT1 (IGN signal) HI.	
	Set IN1(IGN confirmation signal) at HI		
Transfer front wheels onto the roller	Turn on ABS lamp Turn off enter and exit lamp Blink test lamp Turn on front wheel lamp	When there is no response within Max2Sec, it is judged as error on B/T side. (DTC; A1100) IN1=HI ? YES	
	ExtendedSessionEntry/confirmation for the current DTC :DIAGCLR01		
Pedal the brake by 30 kg	Test lamp on Start the roller revolution pedal the brake;lamp on ->normal brake test		
	Brake force test (BRKCHK01)・front right wheel		
	Brake force test (BRKCHK01)・front left wheel		
Brake pedal off	Blink the sensor lamp		
	Wheel speed sensor output test (SPCHK01)・front		
Transfer rear wheels onto the roller Side brake test Pedal the brake by 30 kg	Test lamp on Start the roller revolution pedal the brake;lamp on ->normal brake test		
	Brake force test (BRKCHK01) rear right wheel		
	Brake force test (BRKCHK01) rear left wheel		
Brake pedal off	Blink the sensor lamp		
	Wheel speed sensor output test (SPCHK01)・rear		
	FC memory test (FCCHK01)		
Ignition Off connector removing			

3． Diagﾓｰﾄﾞｴﾝﾄﾘｰ/現在故障有無の確認ﾌﾛｰ：DIAGCLR01

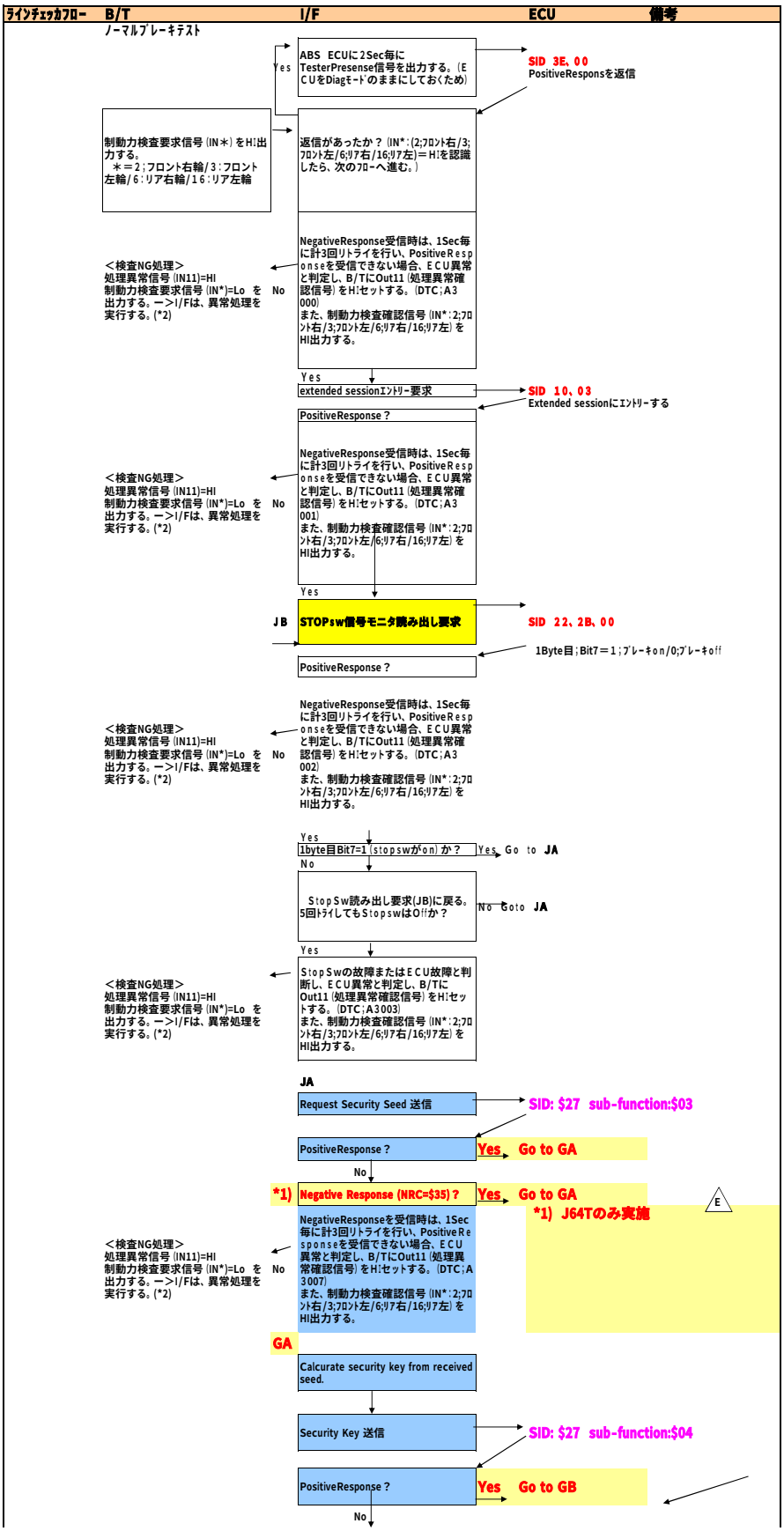


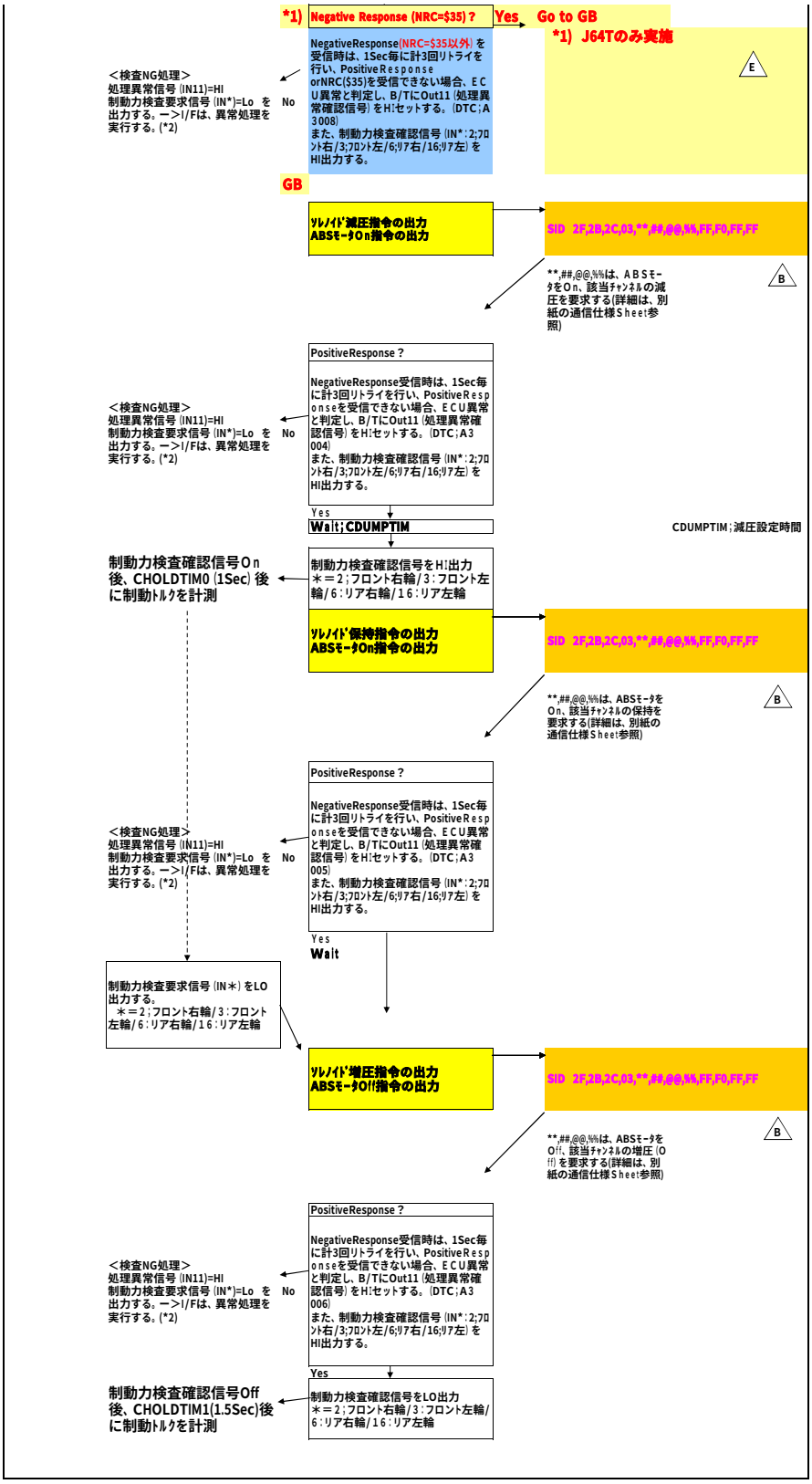
3. ExtemdedSesion Entry/the confirmation for the current DTC check flow:DIAGCLR01

F English description was added.

Line checker flow	B/T	I/F	ECU	Remark
		Extended session entry requirement	SID 10,03 entry to Extended session	
		＊1 PositiveResponse ?		
	NG ! !	No: In case NegativeResponse is received, retry every 1Sec, total of three times.If PositiveResponse can not be received , the I/F detect ECU failure, and set Out11 (process error check signal) HI to B/T (DTC; A2000)		
		Yes		
		Wait; CDIAGINT2(Sec)		CDIAGINT2＝2SEC
		DTC clear requirement	SID 14,FF,FF,FF clear DTC	
	NG ! !	No: In case NegativeResponse is received, retry every 1Sec, total of three times.If PositiveResponse can not be received , the I/F detect ECU failure, and set Out11 (process error check signal) HI to B/T (DTC; A2007) (note) If NegativeResponse;78 is received,the I/F shall stay waiting.		
		Yes		
		Wait; CDIAGINT1(Sec)		CDIAGINT1＝3SEC(J64F)
		DTC reading requirement	SID 19,02,02 Read DTC	
	NG ! !	No: In case NegativeResponse is received, retry every 1Sec, total of three times.If PositiveResponse can not be received , the I/F detect ECU failure, and set Out11 (process error check signal) HI to B/T (DTC; A2004) (note) If NegativeResponse;78 is received,the I/F shall stay waiting.		
		Yes		
		If the current DTC exist, it shall be assessed as NG, and display DTC on the screen. Is there DTC?	No	
		YES		
		Display of DTC Output Out12 (FC occurrence check signal) .		
	Output IN12 (FC occurrence signal) at HI.			
		Turn off the IGN signal.		
	Turn off the IGN check signal			
		Turn off coupler connection signal. Turn off Out12 (FC occurrence check signal)	IGN OFF	
	Coupler connection check signal OFF Turn off the FC occurrence signal			
		RETURN		

4. 制動力検査(BRKCHK01)





＊2;制動力検査確認信号(IN*:2;フロント右/3;フロント左/6;リア右/16;リア左)をLo出力する。Out11(処理異常確認信号)をLoに戻す。(B/T側は、Out11信号=Loを認識後にIN11信号をLoにもどす。)

B/T側が再度本検査項目のトライをリセットSWによって実施する可能性がある。この時、I/Fは、本フローの先頭に戻ってフローを実行すること。

4. Brake force test (BRKCHK01)

F

English description was added.

Line check	B/T	I/F	ECU	Remark

Normal Brake Test

Output the brake force test requirement signal (IN＊) at HI.
＊=2; front right wheel/3:front left wheel/6: rear right wheel/16: rear left wheel

Output TesterPresense signal to ABS ECU every 2Sec. (To keep ECU in Extended Session)

Yes

SID 3E,00
return PositiveResponse

Is there response? (After IN*:(2; front right/3; front left/6; rear right/16; rear left)=HI is confirmed, go to next step in the flow.)

In case NegativeResponse is received, retry every 1sec, total of three times.If PositiveResponse can not be received , the I/F detect ECU failure, and set Out11 (process error check signal) HI to B/T (DTC; A3000)
Also, output the brake force test check signal (IN*:2; front right/3; front left/6; rear right /16; rear left) at HI.

No

<Test NG process>
Process error signal (IN11)=HI
Output the brake force test requirement signal (IN*)=Lo—>
I/F executes error process (*2)

Yes

extended session entry requirement

SID 10,03
entry to Extended session

＊1 PositiveResponse ?

In case NegativeResponse is received, retry every 1sec, total of three times.If PositiveResponse can not be received , the I/F detect ECU failure, and set Out11 (process error check signal) HI to B/T (DTC; A3001)
Also, output the brake force test check signal (IN*:2; front right/3; front left/6; rear right /16; rear left) at HI.

No

<Test NG process>
Process error signal (IN11)=HI
Output the brake force test requirement signal (IN*)=Lo—>
I/F executes error process (*2)

Yes

JB

requirement for reading of STOP sw signal data

SID 22,2B,00

1st Byte; Bit7=1; brake on/0; brake off

＊1 PositiveResponse ?

In case NegativeResponse is received, retry every 1sec, total of three times.If PositiveResponse can not be received , the I/F detect ECU failure, and set Out11 (process error check signal) HI to B/T (DTC; A3002)
Also, output the brake force test check signal (IN*:2; front right/3; front left/6; rear right /16; rear left) at HI.

No

<Test NG process>
Process error signal (IN11)=HI
Output the brake force test requirement signal (IN*)=Lo—>
I/F executes error process (*2)

Yes

Bit7 of 1st byte=1 (stop sw is on) ?

Yes Go to JA

No

Return to "requirement for reading of STOP sw signal data".
Does Stop sw stay "off" after 5 trials ?

No Goto JB

Yes

It is assessed as StopSw error or ECU error, and judged as ECU defect; set Out11 (process error check signal) to B/T at HI. (DTC; A3003)
Also, output brake force test check signal (IN*:2; front right/3; front left/6; rear right/16; rear left) at HI.

<Test NG process>
Process error signal (IN11)=HI
Output the brake force test requirement signal (IN*)=Lo—>
I/F executes error process (*2)

JA

Request Security Seed 送信

SID: $27 sub-function:$03

PositiveResponse ?

Yes Go to GA

No

*1) Negative Response (NRC=$35) ?

Yes Go to GA

E

*1) Apply for J64T only.

In case NegativeResponse is received, retry every 1sec, total of three times.If PositiveResponse can not be received , the I/F detect ECU failure, and set Out11 (process error check signal) HI to B/T (DTC; A3002)
Also, output the brake force test check signal (IN*:2; front right/3; front left/6; rear right /16; rear left) at HI.

No

<Test NG process>
Process error signal (IN11)=HI
Output the brake force test requirement signal (IN*)=Lo—>
I/F executes error process (*2)

GA

Calcurate security key from received seed.

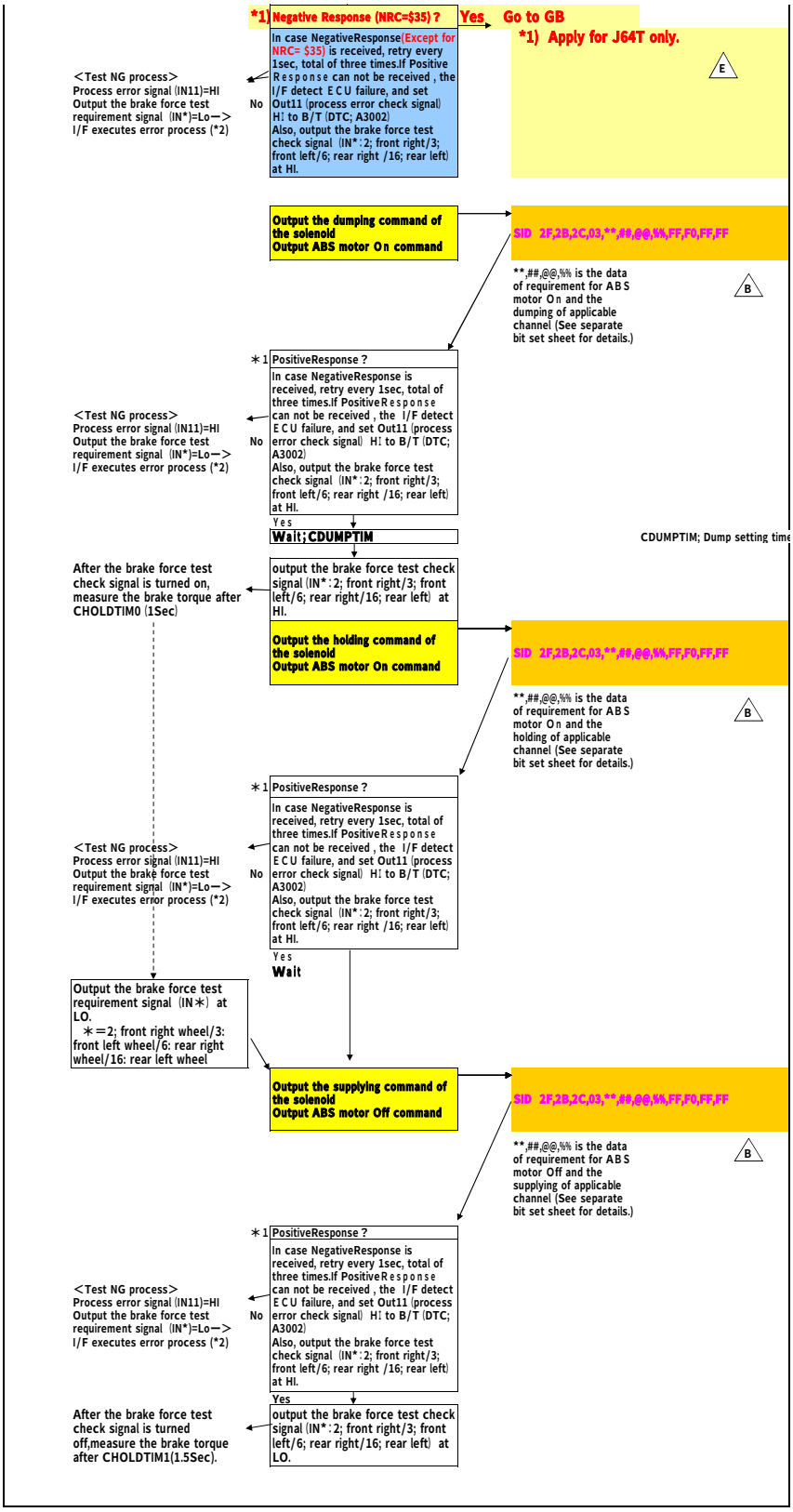
Send to Security Key

SID: $27 sub-function:$04

PositiveResponse ?

Yes Go to GB

No



2; Output the brake force test check signal (IN: 2;front right/3;front left/6; rear right/16;rear left) at Lo. Return Out11 (process error check signal) to Lo. (After B/T recognize Out11 signal at Lo, return IN11 signal to Lo.)
There is a possibility that B/T try to execute this test again by the Reset Switch. In this case, I/F comes back to the top of this flow and executes the flow.

5. 車輪速ｾﾝｻ出力検査(SPCHK01)

ﾗｲﾝﾁｪｯｸﾌﾛｰ	B/T	I/F	ECU	備考
	ｾﾝｻ検査			
		ABS ECUに2Sec毎にTesterPresense信号を出力する。(ECUをDiagﾓｰﾄﾞのままにしておくため)	SID 3E、00 PositiveResponseを返信	
＜検査NG処理＞ 処理異常信号(IN11)=HI ー＞I/Fは、異常処理を実行する。(*2)		NegativeResponse受信時は、1Sec毎に計3回リトライを行い、PositiveResponseを受信できない場合、ECU異常と判定し、B/TにOut11(処理異常確認信号)をHIセットする。(DTC;A4000)		
	ｵﾌﾞｼｮﾝ	STOPsw信号モニタ読み出し要求	SID 22、2B、00 1Byte目;Bit7＝1;ﾌﾞﾚｰｷon/0;ﾌﾞﾚｰｷoff	
	＊1	PositiveResponse？		
＜検査NG処理＞ 処理異常信号(IN11)=HI ｾﾝｻ検査要求信号(IN*)=Lo を出力する。ー＞I/Fは、異常処理を実行する。(*2)	No	NegativeResponse受信時は、1Sec毎に計3回リトライを行い、PositiveResponseを受信できない場合、ECU異常と判定し、B/TにOut11(処理異常確認信号)をHIセットする。(DTC;A4001) また、ｾﾝｻ検査確認信号(IN*:4;ﾌﾛﾝﾄ右/5;ﾌﾛﾝﾄ左/7;ﾘｱ右/8;ﾘｱ左)をHI出力する。		
		1byte目Bit7=0(stopswがOFF)か？	Yes Go to JA	
		StopSw読み出し要求に戻る。5回ﾄﾗｲしてもStopswはOnか？	No Go to JB	
＜検査NG処理＞ 処理異常信号(IN11)=HI ｾﾝｻ検査要求信号(IN*)=Lo を出力する。ー＞I/Fは、異常処理を実行する。(*2)		StopSwの故障またはECU故障と判断し、ECU異常と判定し、B/TにOut11(処理異常確認信号)をHIセットする。(DTC;A4002) また、ｾﾝｻ検査確認信号(IN*:4;ﾌﾛﾝﾄ右/5;ﾌﾛﾝﾄ左/7;ﾘｱ右/8;ﾘｱ左)をHI出力する。		
	JA			
ｾﾝｻ検査要求信号(IN＊)をHI出力する。 ＊＝4;フロント右輪/5:フロント左輪/7:リア右輪/8:リア左輪		IN＊＝HIを認識したら、次のﾌﾛｰへ進む。		
		ｾﾝｻ検査要求信号(OUT＊)をHI出力する。 ＊＝4;フロント右輪/5:フロント左輪/7:リア右輪/8:リア左輪		
	JD	ｾﾝｻｽﾋﾟｰﾄﾞ読み出し	SID 22、$@@,$&&	
			:@@,&&:読み出し対象車輪ID 2B06;FLｽﾋﾟｰﾄﾞ(km/h)、 2B07;FRｽﾋﾟｰﾄﾞ(km/h)、 2B08;RLｽﾋﾟｰﾄﾞ(km/h)、 2B09;RRｽﾋﾟｰﾄﾞ(km/h)	
	＊1	PositiveResponse？		
＜検査NG処理＞ 処理異常信号(IN11)=HI ー＞I/F は、異常処理を実行する。(*2)	No	NegativeResponse受信時は、1Sec毎に計3回リトライを行い、PositiveResponseを受信できない場合、ECU異常と判定し、B/TにOut11(処理異常確認信号)をHIセットする。(DTC;A4003)		
		IN*=4or5(ﾌﾛﾝﾄ)の時、 B＞FLｽﾋﾟｰﾄﾞ≧A(km/h) B＞FRｽﾋﾟｰﾄﾞ≧A(km/h) C≦RLｽﾋﾟｰﾄﾞ＜D(km/h) C≦RRｽﾋﾟｰﾄﾞ＜D(km/h) か？	Yes Go to JC	A: 3.0km/h B:8.0km/h C: 0km/h D:2.0km/h
		IN*=7or8(ﾘｱ)の時、 C≦FLｽﾋﾟｰﾄﾞ＜D(km/h) C≦FRｽﾋﾟｰﾄﾞ＜D(km/h) B＞RLｽﾋﾟｰﾄﾞ≧A(km/h) B＞RRｽﾋﾟｰﾄﾞ≧A(km/h) か？	Yes Go to JC	
		ｾﾝｻｽﾋﾟｰﾄﾞ読み出し要求に戻る。5回ﾄﾗｲしてもNoとなるか？	No Go to JD	
＜検査NG処理＞ 処理異常信号(IN11)=HI ー＞I/F は、異常処理を実行する。(*2)		ｾﾝｻ信号の故障またはECU故障と判断し、B/TにOut11(処理異常確認信号)をHIセットする。(DTC;A4004)		
	JC			
		＜ｾﾝｻ信号確認結果;OK＞ OUT10(処理正常確認信号)をHIセットする。		
IN10(処理正常信号)＝HI出力する。ｾﾝｻ検査要求信号(IN＊)をLO出力する。 ＊＝4;フロント右輪/5:フロント左輪/7:リア右輪/8:リア左輪				
		･ｾﾝｻ検査要求信号(OUT＊)をLO出力する。 ＊＝4;フロント右輪/5:フロント左輪/7:リア右輪/8:リア左輪 ･OUT10(処理正常確認信号)をLOセットする。		
IN10(処理正常信号)=LO出力する。				

＊2；ｾﾝｻ検査要求信号をＬｏ出力する。Out11（処理異常確認信号）をＬｏに戻す。（B/T側は、Out11信号=Loを認識後にIN11信号をLoにもどす。）
　　B/T側が再度本検査項目のﾄﾗｲをリセットＳＷによって実施する可能性がある。この時、I/Fは、本ﾌﾛｰの先頭に戻ってﾌﾛｰを実行すること。

5. Wheel speed sensor output test (SPCHK01)

F English description was added.

Line checker : B/T	I/F	ECU	備考

Sensor test

Output TesterPresense signal every 2sec to ABS ECU. (To keep ECU in Extended session) → SID 3E,00 return PositiveResponse

In case NegativeResponse is received, retry every 1sec, total of three times.If PositiveResponse can not be received , the I/F detect ECU failure, and set Out11 (process error check signal) HI to B/T (DTC; A4000)

<Test NG process>
Process error (IN11)=HI -> The I/F executes error handling (*2)

Yes

JB requirement for reading of STOP sw signal data → SID 22,2B,00

1st Byte; Bit7=1; brake on/0; brake off

*1 PositiveResponse ?

In case NegativeResponse is received, retry every 1sec, total of three times.If PositiveResponse can not be received , the I/F detect ECU failure, and set Out11 (process error check signal) HI to B/T (DTC; A4001)
Also, output the sensor test check signal (IN*: 4; front right/5; front left/7; rear right/8; rear left) at HI.

<Test NG process>
Process error signal (IN11)=HI
Output the sensor test requirement signal (IN*)=Lo. -> The I/F executes error handling (*2)

No

Yes

Bit7 of 1st byte=0 (stop sw is off) ? Yes Go to JA

No

Return to requirement for reading of STOP sw signal data".
Does Stop sw stay "on" after 5 trials ? No Go to JB

Yes

It is assessed as StopSw error or ECU error, and judged as ECU defect; set Out11 (process error check signal) to B/T at HI. (DTC; A4002)
Also, output the sensor test check signal (IN*: 4; front right/5; front left/7; rear right/8; rear left) at HI.

<Test NG process>
Process error signal (IN11)=HI
Output the sensor test requirement signal (IN*)=Lo. -> The I/F executes error handling (*2)

JA

Output the sensor test requirement signal (IN*) at HI.
*=4; front right wheel/5: front left wheel/7: rear right wheel/8: rear left wheel

After checking IN*=HI, go to next step in the flow.

Output the sensor test requirement signal (OUT*) at HI.
*=4; front right wheel/5: front left wheel/7: rear right wheel/8: rear left wheel

JD reading of the sensor speed → SID 22,$@@,$&&

:@@,&&the target wheel ID
2B 06;Front Left(km/h),
2B 07;Front Right(km/h),
2B 08;Rear Left(km/h),
2B 09;Rear Right(km/h)

*1 PositiveResponse ?

In case NegativeResponse is received, retry every 1sec, total of three times.If PositiveResponse can not be received , the I/F detect ECU failure, and set Out11 (process error check signal) HI to B/T (DTC; A4003)

<Test NG process>
Process error (IN11)=HI -> The I/F executes error handling (*2)

No

Yes

When IN*=4or5(front) ,
B>FL speed≧A(km/h)
B>FR speed≧A(km/h)
C≦RL speed<D(km/h)
C≦RR speed<D(km/h) ? Yes Go to JC

A: 3.0km/h
B: 8.0km/h
C: 0km/h
D: 2.0km/h

When IN*=7or8(rear) ,
C≦FL speed<D(km/h)
C≦FR speed<D(km/h)
B>RL speed≧A(km/h)
B>RR speed≧A(km/h) ? Yes Go to JC

No

Return to requirement for reading of the sensor speed
Is it still No at 5 times trial? No Go to JD

Yes

the I/F detect the failure of the sensor signal or ECU failure, and set Out11 (process error check signal) HI to B/T (DTC; A4004)

<Test NG process>
Process error (IN11)=HI -> The I/F executes error handling (*2)

JC

<Sensor signal check result; OK>
Set OUT10 (process normal check signal) at HI.

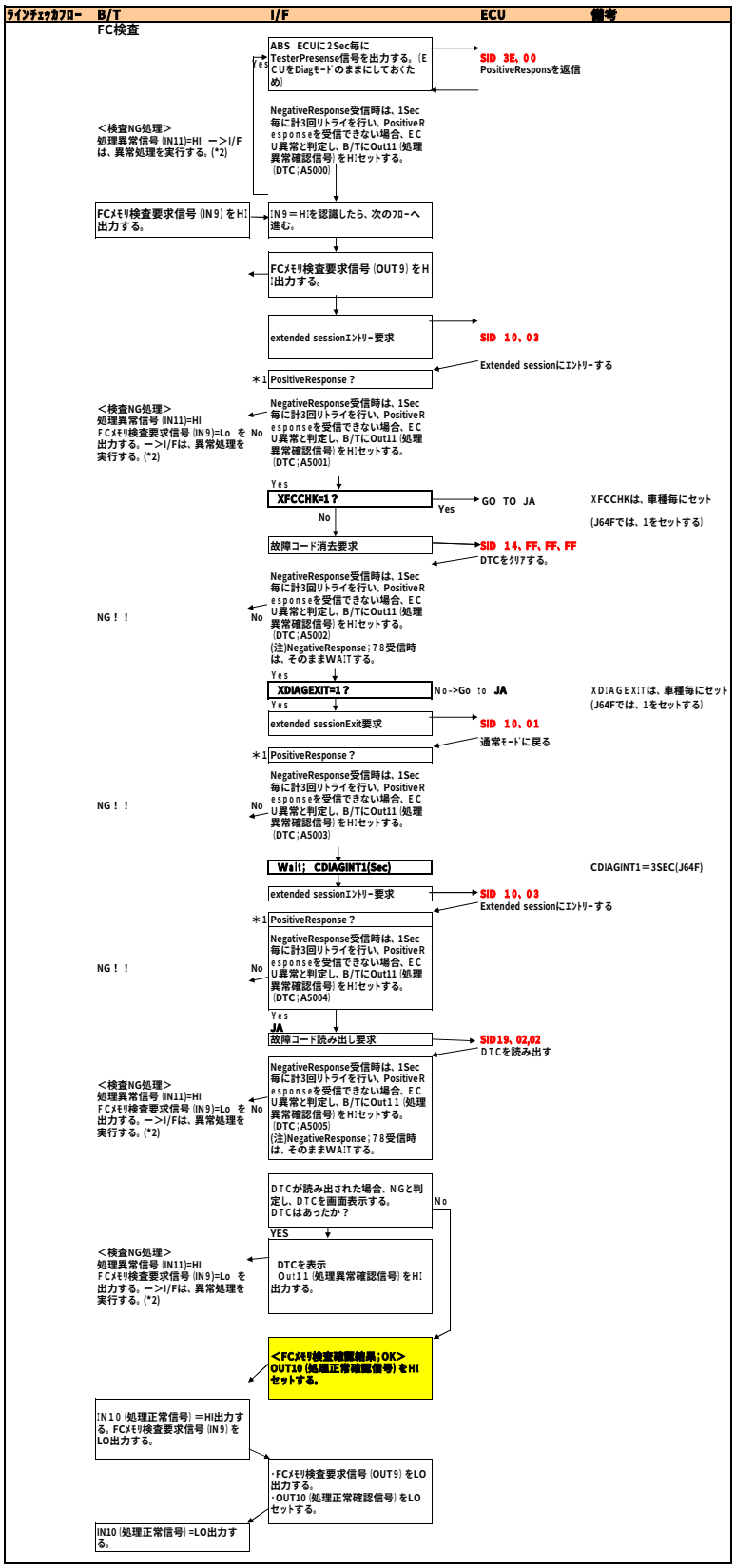
Output IN10 (process normal signal) =HI. Output sensor test requirement signal (IN*) at LO. *=4; front right wheel/5: front left wheel/7: rear right wheel/8: rear left wheel

・Output sensor test requirement signal (OUT*) at LO. *=4; front right wheel/5: front left wheel/7: rear right wheel/8: rear left wheel
・Set OUT10 (process normal check signal) at LO.

Output IN10 (process normal signal) =LO.

	*2; Output the sensor test check signal at Lo. Return Out11 (process error check signal) to Lo. (After B/T recognize Out11 signal at Lo, return IN11 signal to Lo.) There is a possibility that B/T try to execute this test again by the Reset Switch. In this case, I/F comes back to the top of this flow and executes the flow.	

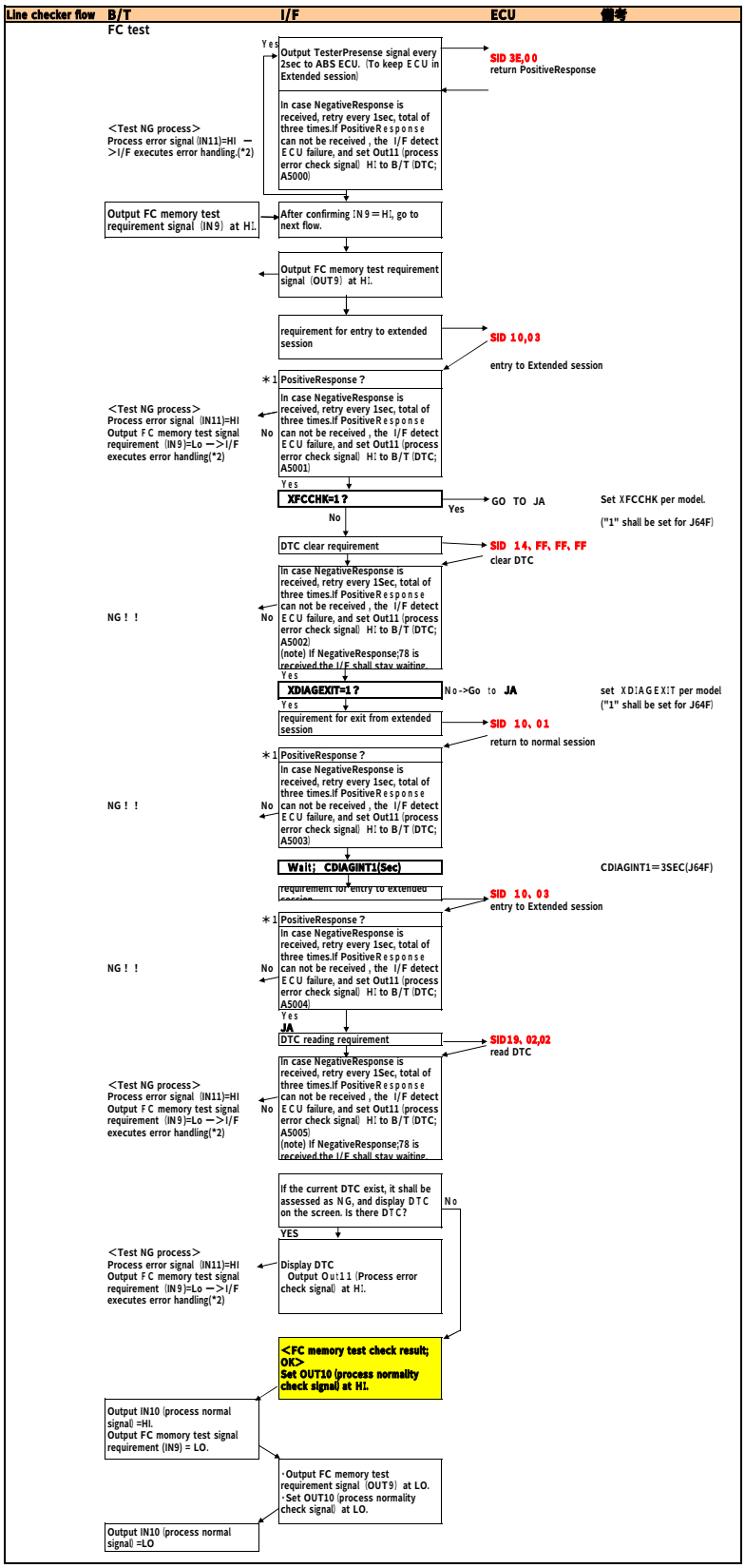
6. FCメモリ検査(FCCHK01)



＊2;FCメモリ検査要求信号をLo出力する。Out11(処理異常確認信号)をLoに戻す。(B/T側は、Out11信号=Loを認識後にIN11信号をLoにもどす。)
B/T側が再度本検査項目のﾄﾗｲをリセットSWによって実施する可能性がある。この時、I/Fは、本ﾌﾛｰの先頭に戻ってﾌﾛｰを実行すること。

6. FC memory test (FCCHK01)

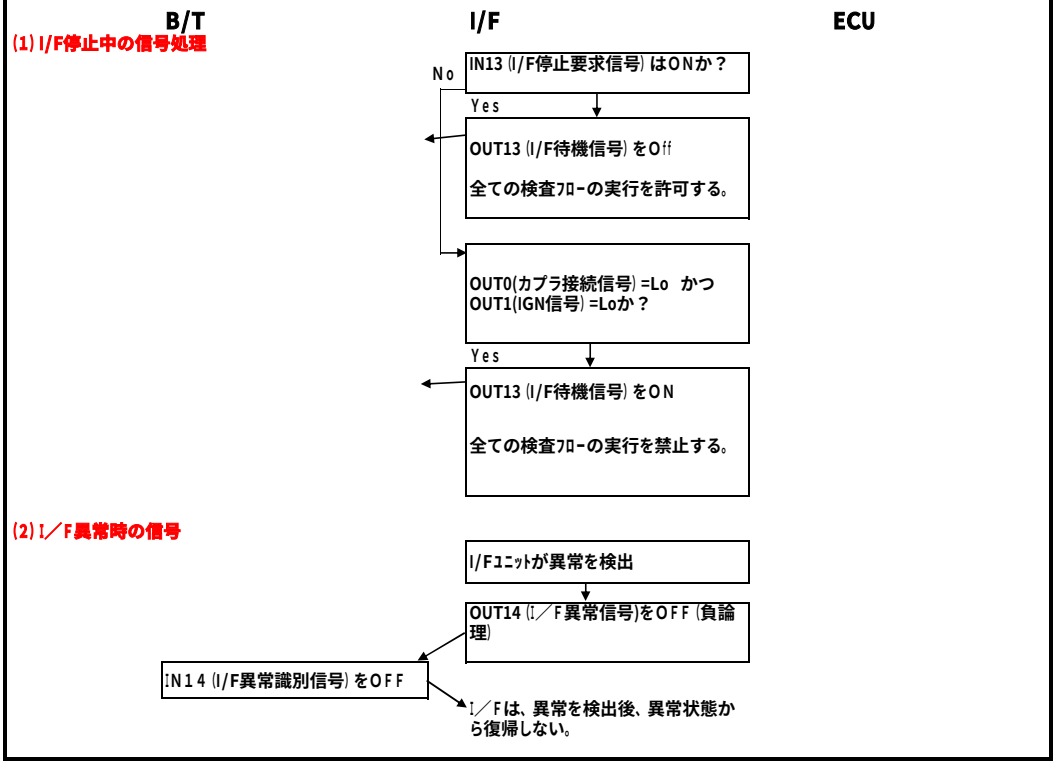
F English description was added



*2; Output the sensor test check signal at Lo. Return Out11 (process error check signal) to Lo. (After B/T recognize Out11 signal at Lo, return IN11 signal to Lo.)
There is a possibility that B/T try to execute this test again by the Reset Switch. In this case, I/F comes back to the top of this flow and executes the flow.

7． 通信異常監視ﾌﾛｰ

本処理は、ﾗｲﾝﾁｪｯｶ実施中には100ms毎に常時、信号の監視を行い、異常時のﾊﾝﾄﾞﾘﾝｸﾞを実行する。



7. Communication error monitor flow

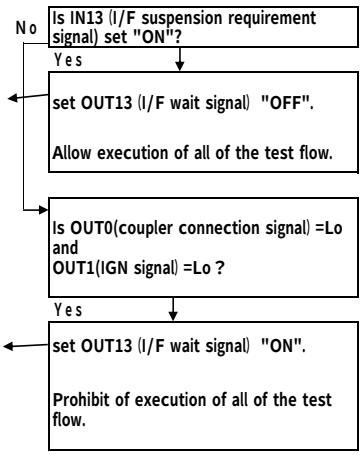
F

English description was added.

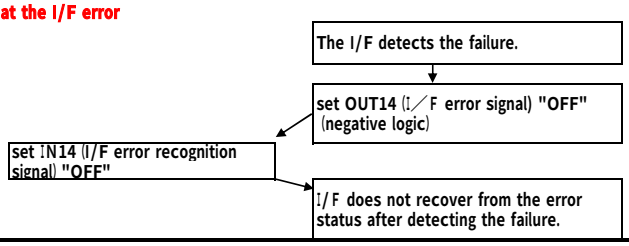
This process always monitors signals and handles errors every 100ms during operation of line checker.

B/T | **I/F** | **ECU**

(1) Signal handling during stopping of the I/F



(2) Signal handling at the I/F error



8. 改定履歴 Change History

改正回数 Revision Number of times	改正区分 Revision Version	年月日 (制改廃状況) Date	改正内容 Content of Change		起案者 Author
			主な改正内容 Change Summary	理由 Reason	
0	A	2008/03/07	新設 New Release.	GGDS標準仕様新設 To apply GGDS standard.	尾中
1	B	2008/06/02	ControlMaskﾃﾞｰﾀ修正 Corrected data of ControlMask.	ﾃﾞｰﾀ設定変更要のため To change setting data.	尾中
2	C	2009/05/29	J74G DSC追加 Added J74G DSC.	車種(ｼｽﾃﾑ)追加 To add project(system).	尾中
3	D	2010/04/12	J7KN DSC追加 Added J7KN DSC.	車種(ｼｽﾃﾑ)追加 To add project(system).	遠藤
4	E	2010/04/20	J68N/J68CC/J68T/J64T追加 Added J68N/J68CC/J68T/J64T.	車種(ｼｽﾃﾑ)追加 To add project(system).	遠藤
5	F	2010/7/13	English description was added.	For J68CC Nanjing Plant.	N.Endo
6	G	2010/10/25	Delete "Line check execution sheet" and error correction	Delete spec. and error correction	H.Sato

Reference)

The bit setting of $2B2C($data1-$data4) for the brake force inspection of the each wheel

the inspected wheel	Mode	$2B2C $data1,$data2,$data3,$data4
FR (Front Right)	Dump	$0C $00 $01 $00
	Hold	$08 $00 $00 $00
	Supply	$00 $00 $00 $00
FL (Front Left)	Dump	$03 $00 $01 $00
	Hold	$02 $00 $00 $00
	Supply	$00 $00 $00 $00
RR (Rear Right)	Dump	$C0 $00 $01 $00
	Hold	$80 $00 $00 $00
	Supply	$00 $00 $00 $00
RL (Rear Left)	Dump	**$30** $00 $01 $00
	Hold	$20 $00 $00 $00
	Supply	$00 $00 $00 $00

The non-inspected wheel should be set for Suppy mode(Isolation valve=OFF,Dump valve=OFF).

(Reference1)
ABS active comand

ID Number	Identifier Name	size	Byte	Bit	Bit Description	
2B2C	BCM valves and pump motor	32	1	7	Rear Right Isolation Valve	0:off 1:on
				6	Rear Right Dump	0:off 1:on
				5	Rear Left Isolation Valve	0:off 1:on
				4	Rear Left Dump	0:off 1:on
				3	Front Right Isolation Valve	0:off 1:on
				2	Front Right Dump	0:off 1:on
				1	Front Left Isolation Valve	0:off 1:on
				0	Front Left Dump	0:off 1:on
			2	7	always=0	
				6	always=0	
				5	always=0	
				4	always=0	
				3	Front Right /Rear Left Traction Control Isolation valve 2	0:off 1:on
				2	Front Left / Rear Right Traction Control Isolation valve 1	0:off 1:on
				1	Front Right / Rear Left Traction Control Supply 2	0:off 1:on
				0	Front Left / Rear Right Traction Control Supply 1	0:off 1:on
			3	7～2	always=0	
				1	Valve Supply	0:off 1:on
				0	Pump Motor supply	0:off 1:on
			4	7～0	unused(always=0)	